

40.016 The Analytics Edge

Matrix Factorization

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Agenda

- Unsupervised Learning vs. Supervised Learning
- Matrix Factorization
 - Eigendecomposition (leading to PCA)
 - Singular Value Decomposition (SVD)
- SVD Applied to Image Compression
- Principal Component Analysis (PCA)
- PCA Applied to Social Progress Index

Supervised vs Unsupervised Learning

	Supervised Learning	Unsupervised Learning
Data	Data is labelled. Output: y , input: $x_1, \dots, x_p$.	Data is unlabeled. Only input: $x_1, \dots, x_p$.
Goal	Predict the response or classify examples.	Understand data structure. Clustering, dimension reduction.
Assessment	Split into training and test sets to evaluate performance.	Hard to assess; no ground truth labels.
Methods	Linear/logistic regression, decision trees, random forests, etc.	PCA, SVD, k-means, hierarchical clustering, etc.

Matrix Factorization

Matrix Factorization

Why decompose a matrix?

- Complex matrices can be difficult to understand directly.
- Decomposition helps break down transformations into simpler building blocks.
- Identify key information from decomposition.
- Eigendecomposition (square matrix), Singular value decomposition (square or rectangular)

Eigendecomposition

Consider a symmetric $n \times n$ matrix $\mathbf{A}$ with real-valued entries. We have

$$\mathbf{A} = \mathbf{U} \mathbf{\Lambda} \mathbf{U}^\top,$$

where

$$\mathbf{A} = \underbrace{\begin{bmatrix} \cdots & \mathbf{u}_j & \cdots \end{bmatrix}}_{\mathbf{U}} \underbrace{\begin{bmatrix} \lambda_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \lambda_n \end{bmatrix}}_{\mathbf{\Lambda}} \underbrace{\begin{bmatrix} \vdots \\ \mathbf{u}_j^\top \\ \vdots \end{bmatrix}}_{\mathbf{U}^\top}$$

- $\mathbf{A} \mathbf{u}_j = \lambda_j \mathbf{u}_j$ for all $j = 1, \dots, n$, where $\mathbf{u}_j \in \mathbb{R}^n$
- $\mathbf{u}_j^\top \mathbf{u}_j = 1$ (or equivalently $\|\mathbf{u}_j\| = 1$) for all $j = 1, \dots, n$
- orthogonal: $\mathbf{u}_i^\top \mathbf{u}_j = 0$ for $i \neq j$

Following convention, let $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n$.

Eigendecomposition

Example. $\mathbf{A} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^\top$

$$\mathbf{U} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}, \quad \mathbf{\Lambda} = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}, \quad \mathbf{A} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^\top = \begin{bmatrix} 1.5 & 0.5 \\ 0.5 & 1.5 \end{bmatrix}.$$

What does $\mathbf{A}$ do to a shape? rotation $\rightarrow$ scaling $\rightarrow$ rotation back

- a rotation matrix $\mathbf{U}^\top$
- a scaling matrix $\mathbf{\Lambda}$
- a rotation back matrix $\mathbf{U}$

Let's apply $\mathbf{A}$ to a unit circle (next slide).

Eigendecomposition

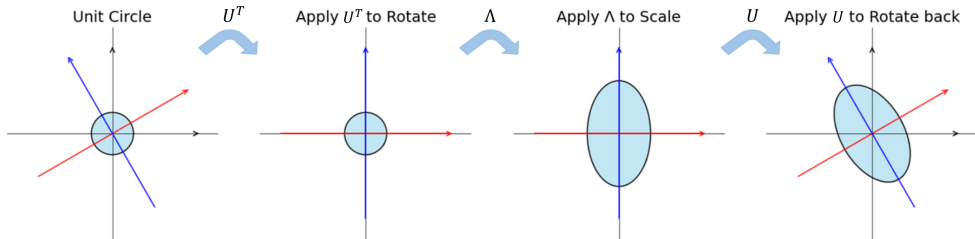


Figure: Eigen-decomposition visualization (Source: Google Image)

Eigendecomposition

Key takeaway. A symmetric matrix transforms space in three steps:

- Pick special directions (transpose of eigenvectors)
- Scaling (eigenvalues)
- Rotate back (eigenvectors)

Eigendecomposition

An Optimization Perspective. What is the unit direction $\mathbf{u}$ for which the quadratic form $\mathbf{u}^\top \mathbf{A} \mathbf{u}$ is maximized? Solving

$$\max_{\|\mathbf{u}\|=1} \mathbf{u}^\top \mathbf{A} \mathbf{u}.$$

Eigendecomposition

$$\begin{aligned} \max_{\mathbf{u}} \quad & \mathbf{u}^\top \mathbf{A} \mathbf{u} \\ \text{subject to} \quad & \mathbf{u}^\top \mathbf{u} - 1 = 0. \end{aligned}$$

Using the method of Lagrange multipliers, with the Lagrange multiplier λ , we define the Lagrangian function:

$$\mathcal{L}(\mathbf{u}, \lambda) = \mathbf{u}^\top \mathbf{A} \mathbf{u} - \lambda (\mathbf{u}^\top \mathbf{u} - 1).$$

By the Lagrange multiplier theorem, any optimal solution must satisfy $\frac{\partial \mathcal{L}}{\partial \lambda} = 0$ and $\frac{\partial \mathcal{L}}{\partial \mathbf{u}} = 0$.

Eigendecomposition

Taking partial derivatives, we have

$$\frac{\partial \mathcal{L}}{\partial \lambda} = -\mathbf{u}^\top \mathbf{u} + 1, \quad \frac{\partial \mathcal{L}}{\partial \mathbf{u}} = 2\mathbf{A}\mathbf{u} - 2\lambda\mathbf{u}.$$

By equating the above to 0:

$$\mathbf{u}^\top \mathbf{u} = 1, \quad \mathbf{A}\mathbf{u} = \lambda\mathbf{u}.$$

- The optimal value λ is an eigenvalue of $\mathbf{A}$ and $\mathbf{u}$ is its associated eigenvector.
- The maximum value of $\mathbf{u}^\top \mathbf{A}\mathbf{u} = \mathbf{u}^\top \lambda\mathbf{u} = \lambda$ is attained for $\mathbf{u} = \mathbf{u}_1$ and the value is λ_1 (i.e., the maximum eigenvalue with the corresponding eigenvector).

Eigendecomposition

- The quadratic form $\mathbf{u}^\top \mathbf{A} \mathbf{u}$ is maximized along $\mathbf{u}_1$, and its maximum value is the largest eigenvalue λ_1 .
- What is the second meaningful direction (different from what $\mathbf{u}_1$ already captures)?
Solving

$$\max_{\|\mathbf{u}\|=1, \mathbf{u} \perp \mathbf{u}_1} \mathbf{u}^\top \mathbf{A} \mathbf{u}.$$

- The optimal solution is attained for $\mathbf{u} = \mathbf{u}_2$ and the optimal value is λ_2 (i.e., the second largest eigenvalue with the corresponding eigenvector).
- We can continue like this.

Singular Value Decomposition

Consider an $m \times n$ matrix $\mathbf{A}$ with real-valued entries. We have

$$\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$$

where

$$\underbrace{\begin{bmatrix} \mathbf{A} \end{bmatrix}}_{m \times n} = \underbrace{\begin{bmatrix} \mathbf{u}_1, \dots, \mathbf{u}_m \end{bmatrix}}_{m \times m} \underbrace{\begin{bmatrix} \sigma_1 & & 0 & \dots \\ & \ddots & 0 & \dots \\ & & \sigma_r & 0 & \dots \\ 0 & \dots & 0 & \dots \end{bmatrix}}_{\mathbf{\Sigma} \text{ is } m \times n \text{ matrix, } r \leq \min(m, n)} \underbrace{\begin{bmatrix} \mathbf{v}_1^\top \\ \vdots \\ \mathbf{v}_n^\top \end{bmatrix}}_{n \times n}$$

- $\mathbf{u}_j^\top \mathbf{u}_j = 1$ for $j = 1, \dots, m$ and $\mathbf{u}_i^\top \mathbf{u}_j = 0$ for $i \neq j$. Each $\mathbf{u}_j \in \mathbb{R}^m$.
- $\mathbf{v}_j^\top \mathbf{v}_j = 1$ for $j = 1, \dots, n$ and $\mathbf{v}_i^\top \mathbf{v}_j = 0$ for $i \neq j$. Each $\mathbf{v}_j \in \mathbb{R}^n$.

Following convention, let $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r > 0$.

Singular Value Decomposition

Example. $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T$,

$$\mathbf{U} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}, \quad \mathbf{\Sigma} = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}, \quad \mathbf{V}^T = \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Multiplying them gives the matrix $\mathbf{A}$:

$$\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T = \begin{bmatrix} -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \\ \frac{3}{\sqrt{2}} & -\frac{3}{\sqrt{2}} & 0 \end{bmatrix}.$$

- $\mathbf{V}^T \in \mathbb{R}^{n \times n}$: rotation in the input space
- $\mathbf{\Sigma} \in \mathbb{R}^{m \times n}$: scaling and possible dimension reduction
- $\mathbf{U} \in \mathbb{R}^{m \times m}$: rotation in the output space

Singular Value Decomposition

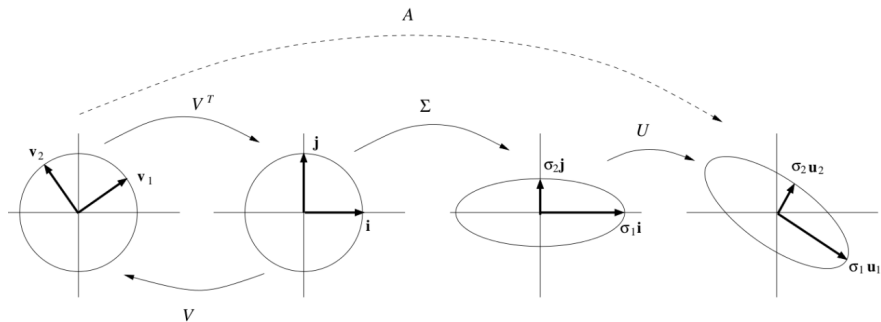


Figure: Singular Value Decomposition Visualization (Source: Google Image)

Singular Value Decomposition

Key Takeaway. Singular Value Decomposition (SVD) expresses any matrix as a sequence of intuitive geometric actions:

1. Rotate the input space via $\mathbf{V}^T$
2. Scale (stretch or compress) along principal directions via $\mathbf{\Sigma}$
3. Rotate the output space via $\mathbf{U}$

Relationship Between Eigendecomposition and SVD

For a matrix $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$, consider the matrix $\mathbf{A}\mathbf{A}^\top$:

$$\begin{aligned}\mathbf{A}\mathbf{A}^\top &= (\mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top) (\mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top)^\top \\ &= \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top \mathbf{V}\mathbf{\Sigma}^\top \mathbf{U}^\top \\ &= \mathbf{U}\mathbf{\Sigma}\mathbf{\Sigma}^\top \mathbf{U}^\top \quad (\text{since } \mathbf{V}^\top \mathbf{V} = \mathbf{I}) \\ &= \mathbf{U} \begin{bmatrix} \sigma_1^2 & \dots & 0 & 0 & 0 \\ \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & \dots & \sigma_r^2 & 0 & 0 \\ 0 & \dots & 0 & 0 & 0 \\ 0 & \dots & 0 & 0 & \ddots \end{bmatrix} \mathbf{U}^\top.\end{aligned}$$

The column vectors in $\mathbf{U}$ are exactly the eigenvectors of $\mathbf{A}\mathbf{A}^\top$. The nonzero eigenvalues of $\mathbf{A}\mathbf{A}^\top$ are $\sigma_1^2, \dots, \sigma_r^2$, and the remaining eigenvalues, if any, are zero.

Recall that $r \leq \min(m, n)$.

Relationship Between Eigendecomposition and SVD

For a matrix $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$, consider the matrix $\mathbf{A}^\top \mathbf{A}$:

$$\begin{aligned}\mathbf{A}^\top \mathbf{A} &= (\mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top)^\top (\mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top) \\ &= \mathbf{V}\mathbf{\Sigma}^\top \mathbf{U}^\top \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top \\ &= \mathbf{V}\mathbf{\Sigma}^\top \mathbf{\Sigma}\mathbf{V}^\top \quad (\text{since } \mathbf{U}^\top \mathbf{U} = \mathbf{I}) \\ &= \mathbf{V} \begin{bmatrix} \sigma_1^2 & \dots & 0 & 0 & 0 \\ \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & \dots & \sigma_r^2 & 0 & 0 \\ 0 & \dots & 0 & 0 & 0 \\ 0 & \dots & 0 & 0 & \ddots \end{bmatrix} \mathbf{V}^\top.\end{aligned}$$

The column vectors in $\mathbf{V}$ are exactly the eigenvectors of $\mathbf{A}^\top \mathbf{A}$. The nonzero eigenvalues of $\mathbf{A}^\top \mathbf{A}$ are $\sigma_1^2, \dots, \sigma_r^2$, and the remaining eigenvalues, if any, are zero.

SVD Applied to Image Compression

What's the Difference?



What's the Difference?



Low-Rank Approximation with SVD

Why Low-Rank Approximation?

1. Reduce the storage space
2. A low-rank approximation is perhaps a natural representation of reality

Image Sharing

- Image sharing: Whatsapp, Snapchat, Facebook, Instagram, etc.
- ~ 14 billion images are shared per day (2024). Used to be ~ 1.8 billion in 2014.
- Images are not shared at best quality. But, compression needs to maintain
 - minimum quality
 - user experience

Image Storage and Compression

- A digital image: a finite number of pixels arranged as a matrix.
 - Grayscale images contain a value representing the light intensity from 0 to 1 at each pixel.
 - Color images may have a set of stacked colors depending on format: RGB, CMYK
- Use matrix factorization for image compression
 - Singular value decomposition

Low-rank Approximation with SVD

Idea

Develop a low-rank approximation of $\mathbf{A}$ by discarding the smaller singular values and the corresponding singular vectors.

Low-rank Approximation with SVD

$$\underbrace{[\mathbf{A}]}_{m \times n} = \underbrace{[\mathbf{U}]}_{m \times m} \underbrace{\begin{bmatrix} \sigma_1 & & 0 & \dots \\ & \ddots & & \\ & & \sigma_r & 0 & \dots \\ 0 & & \dots & \dots \end{bmatrix}}_{\Sigma \text{ is } m \times n \text{ matrix}} \underbrace{[\mathbf{V}^\top]}_{n \times n}$$

A **rank- k approximation** $\hat{\mathbf{A}}$ of $\mathbf{A}$ where $k \leq r$ (here r is the rank of $\mathbf{A}$) is given by:

$$\underbrace{[\hat{\mathbf{A}}]}_{m \times n} = \underbrace{[\hat{\mathbf{U}}]}_{m \times k} \underbrace{\begin{bmatrix} \sigma_1 & & \\ & \ddots & \\ & & \sigma_k \end{bmatrix}}_{\hat{\Sigma} \text{ is } k \times k \text{ matrix}} \underbrace{[\hat{\mathbf{V}}^\top]}_{k \times n}$$

where $\sigma_1 \geq \dots \geq \sigma_k > 0$ are the k largest singular values of $\mathbf{A}$.

We use *only the first k columns* of $\mathbf{U}$ and $\mathbf{V}$ to create $\hat{\mathbf{U}}$ and $\hat{\mathbf{V}}$, instead of all columns.

Low-rank Approximation with SVD

Reduce the storage space:

- Storage for the original $m \times n$ matrix: mn
- Storage for its rank- k approximation: $mk + k + nk$

Low-rank Approximation with SVD

Frobenius norm:

$$\|\mathbf{A}\|_F = \sqrt{\sum_{i=1}^m \sum_{j=1}^n A_{ij}^2}$$

When approximating a matrix by a low-rank matrix, the proportion of information (variance) retained by the rank- k approximation can be measured by:

$$\frac{\|\hat{\mathbf{A}}\|_F^2}{\|\mathbf{A}\|_F^2} = \frac{\sum_{i=1}^k \sigma_i^2}{\sum_{i=1}^r \sigma_i^2}.$$

Why? Next slide.

Low-rank Approximation with SVD

$$\begin{aligned}\|\mathbf{A}\|_F^2 &= \sum_{i=1}^m \sum_{j=1}^n A_{ij}^2 \stackrel{(a)}{=} \text{trace}(\mathbf{A}\mathbf{A}^\top) \\ &= \text{trace}\left(\left(\mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top\right)\left(\mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top\right)^\top\right) \\ &= \text{trace}\left(\mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top\mathbf{V}\mathbf{\Sigma}^\top\mathbf{U}^\top\right) \\ &= \text{trace}\left(\mathbf{U}\mathbf{\Sigma}\mathbf{\Sigma}^\top\mathbf{U}^\top\right) \\ &= \text{trace}\left(\mathbf{U}^\top\mathbf{U}\mathbf{\Sigma}\mathbf{\Sigma}^\top\right) \\ &= \text{trace}\left(\mathbf{\Sigma}\mathbf{\Sigma}^\top\right) = \sum_{i=1}^r \sigma_i^2\end{aligned}$$

Low-rank Approximation with SVD

In our previous slide, the equality (a) comes from:

$$\mathbf{A} = \begin{bmatrix} A_{11} & \dots & A_{1n} \\ \vdots & \ddots & \vdots \\ A_{m1} & \dots & A_{mn} \end{bmatrix} \in \mathbb{R}^{m \times n}, \quad \mathbf{A}^\top = \begin{bmatrix} A_{11} & \dots & A_{m1} \\ \vdots & \ddots & \vdots \\ A_{1n} & \dots & A_{mn} \end{bmatrix} \in \mathbb{R}^{n \times m}$$

The sum of diagonal entries of $\mathbf{A}\mathbf{A}^\top$ is $\sum_{i=1}^m \sum_{j=1}^n A_{ij}^2$.

Thus, $\text{trace}(\mathbf{A}\mathbf{A}^\top) = \sum_{i=1}^m \sum_{j=1}^n A_{ij}^2$.

R implementation

- We work on a grayscale image and a color image.
- Feel free to use your own jpeg image.

PCA Applied to Social Progress Index

Social Progress Imperative

- Economic progress/growth perhaps does not provide a holistic view of development, prosperity, or well-being of its citizens.
- A relevant question to ask is if a society/nation meets the basic needs of its citizens in terms of
 - healthcare
 - education
 - clean water
 - human rights
 - environmental quality
 - gender equality
 - public/personal safety
 - ...
- Topic is of interest to social scientists, policy makers, governments, businesses, ...
- A global non-profit organization called Social Progress Imperative publishes the Social Progress Index.

Social Progress Index

The Social Progress Index is a well-established measure, published since 2013, that is meant to catalyze improvement and drive action by presenting social outcome data in a useful and reliable way. Composed of multiple dimensions, the Social Progress Index can be used to benchmark success and provide a holistic, transparent, outcome-based measure of a country's well-being that is independent of economic indicators. Policymakers, businesses, and countries' citizens alike can use it to compare their country against others on different facets of social progress, allowing the identification of specific areas of strength or weakness.

- taken from the Methodology summary

[Video](#)

Social Progress Index

Figure 1 / Social Progress Index Component-Level Framework



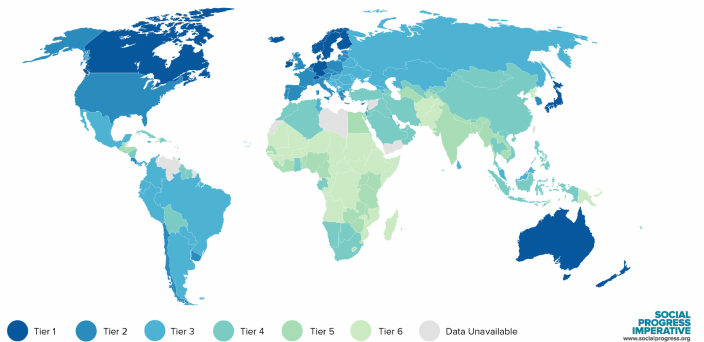
Social Progress Index

Figure 2 / Social Progress Index Indicator-Level Framework



Social Progress Index: Map

2020 Social Progress Index



Key Questions

- What can we understand about social progress from the raw data?
- Can we identify where the major differences in social progress occur?
- Can we propose some kind of social progress index from our analysis?

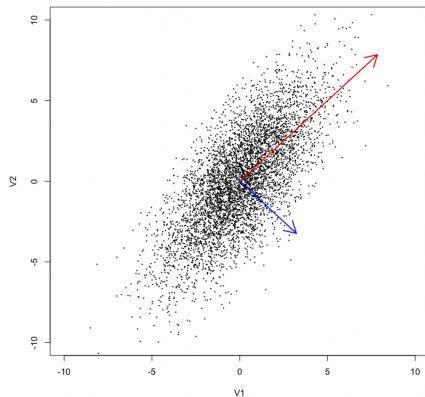
Principal Component Analysis (PCA)

Goal.

- Reduce the dimensionality of the data.
- Find a low-dimensional projection that captures the most variance.
- Discover directions (principal component directions) that best explain variability in the data.

Principal Component Analysis (PCA)

5000 data points from bivariate normal with mean 0, variances 5, 10 and covariance 5.



The red arrow shows the direction of maximum variability $\mathbf{v}_1$, and the blue arrow shows the second orthogonal direction of maximum variability $\mathbf{v}_2$.

Principal Component Analysis (PCA)

Along which direction should we project the data so that the projected data has the largest possible variance?

Formally, let $\mathbf{X} = [X_1, \dots, X_p]^T \in \mathbb{R}^p$ be a random vector with zero mean (i.e., $E[\mathbf{X}] = \mathbf{0}$) and covariance $\mathbf{Q} = E[\mathbf{X}\mathbf{X}^T] \in \mathbb{R}^{p \times p}$.

We want to find a constant unit vector $\mathbf{v}$ such that

$$\max_{\|\mathbf{v}\|=1} \text{Variance} \left(\mathbf{X}^T \mathbf{v} \right).$$

Principal Component Analysis (PCA)

Since $E[\mathbf{X}^\top \mathbf{v}] = E[\mathbf{X}]^\top \mathbf{v} = \mathbf{0}$, we have

$$\text{Variance}(\mathbf{X}^\top \mathbf{v}) = E[(\mathbf{X}^\top \mathbf{v})^2] = E[\mathbf{v}^\top \mathbf{X} \mathbf{X}^\top \mathbf{v}] = \mathbf{v}^\top E[\mathbf{X} \mathbf{X}^\top] \mathbf{v} = \mathbf{v}^\top \mathbf{Q} \mathbf{v},$$

where $\mathbf{Q} = E[\mathbf{X} \mathbf{X}^\top]$.

We solve:

$$\max_{\|\mathbf{v}\|=1} \mathbf{v}^\top \mathbf{Q} \mathbf{v}.$$

As before (Slides 10–12), the solution is the eigenvector $\mathbf{v}_1$ associated with the largest eigenvalue λ_1 of $\mathbf{Q}$.

We call $\mathbf{v}_1$ the first **principal component direction**.

Principal Component Analysis (PCA)

To obtain the second principal component, we solve:

$$\max_{\|\mathbf{v}\|=1, \mathbf{v} \perp \mathbf{v}_1} \mathbf{v}^\top \mathbf{Q} \mathbf{v}.$$

As before (Slide 13), the solution is the eigenvector $\mathbf{v}_2$ associated with the second largest eigenvalue λ_2 of $\mathbf{Q}$.

Continue like this to obtain p principal component directions $\mathbf{v}_1, \dots, \mathbf{v}_p$.

PCA Implementation

Main Idea.

- Compute the first k ($k \leq p$) principal component directions $\mathbf{v}_1, \dots, \mathbf{v}_k$, which are the eigenvectors of covariance $\mathbf{Q} = E[\mathbf{X}\mathbf{X}^\top]$.
- Project the data $\mathbf{x}$ onto the subspace spanned by $\mathbf{v}_1, \dots, \mathbf{v}_k$ to reduce the dimension:

$$\left(\mathbf{x}^\top \mathbf{v}_1\right) \mathbf{v}_1 + \dots + \left(\mathbf{x}^\top \mathbf{v}_k\right) \mathbf{v}_k.$$

- These coefficients $\mathbf{x}^\top \mathbf{v}_i$ are called the **principal component scores**. They tell us how much the data $\mathbf{x}$ aligns with the principal component directions $\mathbf{v}_i$.

PCA Implementation

Data.

1. n : the number of observations
2. p : the number of features
3. For $i = 1, \dots, n$, $\mathbf{x}_i = [x_{i1}, \dots, x_{ip}]^\top \in \mathbb{R}^p$ is the i -th observation with p features.
4. Consider n observations $\mathbf{x}_1, \dots, \mathbf{x}_n$. We observe the $n \times p$ data matrix:

$$\mathbb{X} = \begin{bmatrix} \mathbf{x}_1^\top \\ \vdots \\ \mathbf{x}_n^\top \end{bmatrix} = \begin{bmatrix} x_{11} & \dots & x_{1p} \\ \vdots & \ddots & \vdots \\ x_{n1} & \dots & x_{np} \end{bmatrix}.$$

PCA Implementation

Data Preparation.

1. We center the data so that it has zero sample mean for each feature $j = 1, \dots, p$:

$$\sum_{i=1}^n x_{ij} = 0 \text{ for all } j = 1, \dots, p.$$

2. Denote the sample covariance matrix of data $\mathbf{x}_1, \dots, \mathbf{x}_n$ by

$$\mathbf{S} = \frac{1}{n-1} \sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i^\top = \frac{1}{n-1} \mathbf{X}^\top \mathbf{X},$$

where

- $\mathbf{x}_i \in \mathbb{R}^p$ and $\mathbf{X} \in \mathbb{R}^{n \times p}$.
- $\mathbf{S} \in \mathbb{R}^{p \times p}$ is an unbiased estimator of the true covariance matrix $\mathbf{Q}$ (Slide 42).

PCA Implementation

- We cannot see the true covariance matrix $\mathbf{Q}$ of the population
- We can use the sample covariance matrix $\mathbf{S}$ while doing PCA
- That is, compute the eigendecomposition of $\mathbf{S} = \frac{1}{n-1} \mathbb{X}^\top \mathbb{X}$ to obtain eigenvectors $\mathbf{v}_1, \dots, \mathbf{v}_p$, which are *the principal component directions of data* $\mathbb{X}$.
- We can also use an alternative way: compute the SVD of $\frac{1}{\sqrt{n-1}} \mathbb{X}$.

The eigenvectors of $\mathbf{S} = \frac{1}{n-1} \mathbb{X}^\top \mathbb{X}$ are exactly the right singular vectors of $\frac{1}{\sqrt{n-1}} \mathbb{X}$, because:

- Let $\frac{1}{\sqrt{n-1}} \mathbb{X} = U \Sigma V^\top$.
- Then, $\mathbf{S} = \frac{1}{n-1} \mathbb{X}^\top \mathbb{X} = (U \Sigma V^\top)^\top U \Sigma V^\top = V \Sigma^\top U^\top U \Sigma V^\top = V \Sigma^\top \Sigma V^\top$.

PCA Implementation

Since the data have been centered, i.e.,

$$\sum_{i=1}^n x_{ij} = 0 \text{ for all } j = 1, \dots, p.$$

for variable $X_j = [x_{1j}, \dots, x_{nj}]^\top$,

$$\text{Variance}(X_j) = \frac{1}{n-1} \sum_{i=1}^n (x_{ij} - \bar{x}_j)^2 = \frac{1}{n-1} \sum_{i=1}^n x_{ij}^2,$$

where $\bar{x}_j \stackrel{\text{def}}{=} \frac{1}{n} \sum_{i=1}^n x_{ij} = 0$.

PCA Implementation

Total variance in the data set is:

$$\begin{aligned}\text{Total (sample) variance} &= \text{Variance}(X_1) + \dots + \text{Variance}(X_p) \\ &= \frac{1}{n-1} \sum_{j=1}^p \sum_{i=1}^n x_{ij}^2 \\ &\stackrel{(a)}{=} \text{Trace} \left(\frac{1}{n-1} \mathbb{X}^\top \mathbb{X} \right) \\ &= \text{Trace}(\mathbf{S}) = \text{Trace}(\mathbf{V} \mathbf{\Lambda} \mathbf{V}^\top) \\ &= \text{Trace}(\mathbf{V}^\top \mathbf{V} \mathbf{\Lambda}) = \text{Trace}(\mathbf{\Lambda}) = \sum_{i=1}^p \lambda_i\end{aligned}$$

where $\lambda_1, \dots, \lambda_p$ are the eigenvalues of $\mathbf{S}$, and equality (a) is due to that the sum of diagonal entries of $\mathbb{X}^\top \mathbb{X}$ is $\sum_{j=1}^p \sum_{i=1}^n x_{ij}^2$ (by a similar derivation as in Slide 31).

PCA Implementation

When choosing the first k principal components of $\mathbf{S}$, *the proportion of variance explained by these components* is:

$$R^2 = \frac{\sum_{i=1}^k \lambda_i}{\sum_{i=1}^p \lambda_i}$$

where $\lambda_1, \dots, \lambda_p$ are the eigenvalues of $\mathbf{S}$ sorted from the highest to the lowest.

PCA Implementation

Final Remark:

- Different data features (attributes) have different variances, and thus a PCA gets influenced by the variable with largest variance.
- It is common practice to standardize all variables to have unit variance.
- Then, from Slide 50, the total variance is equal to $\sum_{i=1}^p \lambda_i = p$.
In such a case, it is often recommended to choose the principal component directions as the eigenvectors of $\mathbf{S}$ whose corresponding eigenvalues are greater than 1.

Limitations of PCA

- PCA is sensitive to the scale of the variables (each variable is a feature among observations, e.g., a specific Social Progress Index indicator among all countries). Therefore, variables are often standardized to have zero mean and unit variance before applying PCA.
- Principal components are linear combinations of the original variables and may be difficult to interpret.
- PCA assumes that the main variability in the data can be captured by linear combinations of the variables. It may fail to capture nonlinear structures.

Reference

- Teaching notes.
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- <https://www.datacamp.com/community/tutorials/pca-analysis-r>